

	Relation	Query	Answer	Generation
T-Rex	P19	Francesco Bartolomeo Conti was born in ____.	Florence	Rome [-1.8], Florence [-1.8], Naples [-1.9], Milan [-2.4], Bologna [-2.5]
	P20	Adolphe Adam died in ____.	Paris	Paris [-0.5], London [-3.5], Vienna [-3.6], Berlin [-3.8], Brussels [-4.0]
	P279	English bulldog is a subclass of ____.	dog	dogs [-0.3], breeds [-2.2], dog [-2.4], cattle [-4.3], sheep [-4.5]
	P37	The official language of Mauritius is ____.	English	English [-0.6], French [-0.9], Arabic [-6.2], Tamil [-6.7], Malayalam [-7.0]
	P413	Patrick Oboya plays in ____ position.	midfielder	centre [-2.0], center [-2.2], midfielder [-2.4], forward [-2.4], midfield [-2.7]
	P138	Hamburg Airport is named after ____.	Hamburg	Hess [-7.0], Hermann [-7.1], Schmidt [-7.1], Hamburg [-7.5], Ludwig [-7.5]
	P364	The original language of Mon oncle Benjamin is ____.	French	French [-0.2], Breton [-3.3], English [-3.8], Dutch [-4.2], German [-4.9]
	P54	Dani Alves plays with ____.	Barcelona	Santos [-2.4], Porto [-2.5], Sporting [-3.1], Brazil [-3.3], Portugal [-3.7]
	P106	Paul Toungui is a ____ by profession.	politician	lawyer [-1.1], journalist [-2.4], teacher [-2.7], doctor [-3.0], physician [-3.7]
	P527	Sodium sulfide consists of ____.	sodium	water [-1.2], sulfur [-1.7], sodium [-2.5], zinc [-2.8], salt [-2.9]
	P102	Gordon Scholes is a member of the ____ political party.	Labor	Labour [-1.3], Conservative [-1.6], Green [-2.4], Liberal [-2.9], Labor [-2.9]
	P530	Kenya maintains diplomatic relations with ____.	Uganda	India [-3.0], Uganda [-3.2], Tanzania [-3.5], China [-3.6], Pakistan [-3.6]
	P176	iPod Touch is produced by ____.	Apple	Apple [-1.6], Nokia [-1.7], Sony [-2.0], Samsung [-2.6], Intel [-3.1]
	P30	Bailey Peninsula is located in ____.	Antarctica	Antarctica [-1.4], Bermuda [-2.2], Newfoundland [-2.5], Alaska [-2.7], Canada [-3.1]
	P178	JDK is developed by ____.	Oracle	IBM [-2.0], Intel [-2.3], Microsoft [-2.5], HP [-3.4], Nokia [-3.5]
	P1412	Carl III used to communicate in ____.	Swedish	German [-1.6], Latin [-1.9], French [-2.4], English [-3.0], Spanish [-3.0]
	P17	Sunshine Coast, British Columbia is located in ____.	Canada	Canada [-1.2], Alberta [-2.8], Yukon [-2.9], Labrador [-3.4], Victoria [-3.4]
	P39	Pope Clement VII has the position of ____.	pope	cardinal [-2.4], Pope [-2.5], pope [-2.6], President [-3.1], Chancellor [-3.2]
	P264	Joe Cocker is represented by music label ____.	Capitol	EMI [-2.6], BMG [-2.6], Universal [-2.8], Capitol [-3.2], Columbia [-3.3]
	P276	London Jazz Festival is located in ____.	London	London [-0.3], Greenwich [-3.2], Chelsea [-4.0], Camden [-4.6], Stratford [-4.8]
	P127	Border TV is owned by ____.	ITV	Sky [-3.1], ITV [-3.3], Global [-3.4], Frontier [-4.1], Disney [-4.3]
	P103	The native language of Mammooty is ____.	Malayalam	Malayalam [-0.2], Tamil [-2.1], Telugu [-4.8], English [-5.2], Hindi [-5.6]
	P495	The Sharon Cuneta Show was created in ____.	Philippines	Manila [-3.2], Philippines [-3.6], February [-3.7], December [-3.8], Argentina [-4.0]
ConceptNet	AtLocation	You are likely to find an overflow in a ____.	drain	sewer [-3.1], canal [-3.2], toilet [-3.3], stream [-3.6], drain [-3.6]
	CapableOf	Ravens can ____.	fly	fly [-1.5], fight [-1.8], kill [-2.2], die [-3.2], hunt [-3.4]
	CausesDesire	Joke would make you want to ____.	laugh	cry [-1.7], die [-1.7], laugh [-2.0], vomit [-2.6], scream [-2.6]
	Causes	Sometimes virus causes ____.	infection	disease [-1.2], cancer [-2.0], infection [-2.6], plague [-3.3], fever [-3.4]
	HasA	Birds have ____.	feathers	wings [-1.8], nests [-3.1], feathers [-3.2], died [-3.7], eggs [-3.9]
	HasPrerequisite	Typing requires ____.	speed	patience [-3.5], precision [-3.6], registration [-3.8], accuracy [-4.0], speed [-4.1]
	HasProperty	Time is ____.	finite	short [-1.7], passing [-1.8], precious [-2.9], irrelevant [-3.2], gone [-4.0]
	MotivatedByGoal	You would celebrate because you are ____.	alive	happy [-2.4], human [-3.3], alive [-3.3], young [-3.6], free [-3.9]
	ReceivesAction	Skills can be ____.	taught	acquired [-2.5], useful [-2.5], learned [-2.8], combined [-3.9], varied [-3.9]
	UsedFor	A pond is for ____.	fish	swimming [-1.3], fishing [-1.4], bathing [-2.0], fish [-2.8], recreation [-3.1]

Table 3: Examples of generation for BERT-large. The last column reports the top five tokens generated together with the associated log probability (in square brackets).

that BERT-large is able to recall such knowledge better than its competitors and at a level remarkably competitive with non-neural and supervised alternatives. Note that we did *not* compare the ability of the corresponding architectures and objectives to capture knowledge in a given body of text but rather focused on the knowledge present in the weights of existing pretrained models that are being used as starting points for many researchers’ work. Understanding which aspects of data our commonly-used models and learning algorithms are capturing is a crucial field of research and this paper complements the many studies focused on the learned linguistic properties of the data.

We found that it is non-trivial to extract a knowledge base from text that performs on par to directly using pretrained BERT-large. This is despite providing our relation extraction baseline with only data that is likely expressing target facts, thus reducing potential for false negatives, as well as using a generous entity-linking oracle. We suspected BERT might have an advantage due to the larger amount of data it has processed, so we added Wikitext-103 as additional data to the relation extraction system and observed no significant change in performance. This suggests that while relation extraction performance might be difficult to improve with more data, language mod-

els trained on ever growing corpora might become a viable alternative to traditional knowledge bases extracted from text in the future.

In addition to testing future pretrained language models using the LAMA probe, we are interested in quantifying the variance of recalling factual knowledge with respect to varying natural language templates. Moreover, assessing multi-token answers remains an open challenge for our evaluation setup.

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